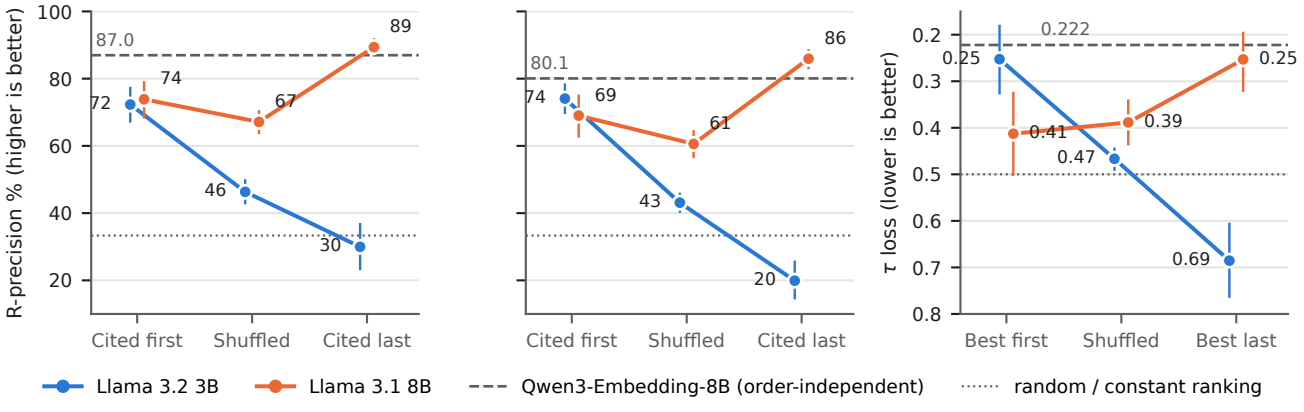
Citation labels, random negatives
Citation labels, concept negatives
Human labels (Stelmakh et al.)
R-precision % (higher is better)
τ loss (lower is better)
87.0
80.1
0.222
Cited first
Shuffled
Cited last
Best first
Best last
Llama 3.2 3B
Llama 3.1 8B
Qwen3-Embedding-8B (order-independent)
random / constant ranking